

An amplitude-level model of π^0 and η electroproduction fitted to CLAS, Hall A and COMPASS

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Abstract

A model of exclusive π^0 and η electroproduction is built directly at the level of the helicity amplitudes, without reference to generalised parton distributions, and fitted to 1555 measured points from CLAS6, CLAS12, Hall A and COMPASS. The purpose is not to test a particular GPD parametrisation but to establish what the data themselves demand of the amplitudes, so that any model can be measured against it. The fit has 28 free parameters and gives $\chi^2/\text{ndf} = 1.4018$ (2140.5/1527). Four conclusions survive a full leave-one-out analysis: the flavour ratio n/p is separated by four neutron points and by nothing else; one polarised-target experiment alone fixes the relative phase of two transverse amplitudes; the preliminary CLAS12 σ_{LT} disagrees in *sign* with all three Hall A measurements; and a single power Q^{n_Q} cannot span Q^2 from 1.2 to 8.3 GeV².

1 The model

1.1 What is parametrised

It is worth being explicit about the level at which this model lives, because three different objects are easily confused.

1. The *generalised parton distributions* themselves, $X^q(x, \xi, t)$ for $X \in \{H_T, \bar{E}_T, \dots\}$ and flavour $q \in \{u, d\}$. These depend on the loop momentum fraction x .
2. The *convolution* of a GPD with the hard subprocess, obtained by integrating x out.
3. The *channel helicity amplitudes*, obtained from the convolutions by combining flavours with the photon couplings and the meson wavefunction.

This model never descends to level 1. No integral over x is performed anywhere, and no GPD is assumed. What is parametrised is the convolution, level 2. That is the whole point of the approach: the data are asked what they require of the convolutions, without committing to a GPD parametrisation that produces them.

The convolution is

$$\langle X \rangle^q(\xi, t, Q^2) = \int_{-1}^1 dx X^q(x, \xi, t) K(x, \xi), \quad K(x, \xi) \sim \frac{1}{x - \xi + i\epsilon}, \quad (1)$$

where K is the hard-scattering kernel. Its precise form is not needed below, because (1) is never evaluated – only parametrised. What matters is a structural consequence of the $i\epsilon$: the convolution is a *complex number*,

$$\text{Im}\langle X \rangle^q = \pi X^q(\xi, \xi, t), \quad \text{Re}\langle X \rangle^q = \mathcal{P} \int dx X^q K, \quad (2)$$

the imaginary part coming from the pole and the real part from the principal value. Since the x -shapes of u and d differ, $\langle X \rangle^u$ and $\langle X \rangle^d$ cannot be assumed to share a phase.

The ansatz. Each convolution is parametrised by a modulus and a phase, separately for each flavour:

$$|\langle X \rangle^q| = N_q \exp[(b_q + b'_q \ln x_B) t] (Q^2)^{n_Q^q/2}, \quad (3)$$

$$\arg \langle X \rangle^d - \arg \langle X \rangle^u = \varphi_X. \quad (4)$$

The overall phase of a channel is unobservable, so u is taken real and only d carries φ_X . The parameters mean:

N	normalisation
b	t -slope extrapolated to $x_B = 1$
b'	the slope at a given x_B is $b + b' \ln x_B$, so $-b'$ plays the role of the Regge α'
n_Q	power of Q^2
φ	phase of the d convolution relative to u

This is the same slope convention as the Goloskokov–Kroll fits. An older parameter convention used $L = \ln x_B - \ln 0.15$, so that b was the slope at $x_B = 0.15$; files in that convention must be converted ($b_{\text{new}} = b + 1.897 b'$), never copied raw.

Three blocks are parametrised this way: $\langle H_T \rangle$ and $\langle \bar{E}_T \rangle$, which build the transverse amplitudes, and a longitudinal block $\langle L \rangle$, which builds $T_{00}^{(-)}$. The labels H_T and $\bar{E}_T$ name the GPD being convolved; the object that carries the parameters is always the convolution, never the GPD itself.

Assumptions, stated plainly. Equations (3) and (4) embody two approximations that should not be hidden. A single exponential in t for the modulus, and a t -independent phase, are together equivalent to assuming that $\text{Re} \langle X \rangle$ and $\text{Im} \langle X \rangle$ share a common t -slope: if they did not, the modulus $\sqrt{\text{Re}^2 + \text{Im}^2}$ would not be a single exponential and $\varphi = \arctan(\text{Im} / \text{Re})$ would depend on t . Nothing forces that common slope; it is an assumption of convenience, and relaxing it costs one parameter per block ($\varphi \rightarrow \varphi_0 + \varphi_1 t$, as already used for the phase of $T_{00}^{(-)}$).

1.2 From convolutions to channel amplitudes

The observables are built from six channel helicity amplitudes. In the notation of the π^0 amplitude note,

$T_{00}^{(+)}$	longitudinal, recoil non-flip	$T_{00}^{(-)}$	longitudinal, recoil flip
$T_{01}^{(+)}$	natural transverse, non-flip	$T_{01}^{(-)}$	natural transverse, flip
$U_{01}^{(+)}$	unnatural transverse, non-flip	$U_{01}^{(-)}$	unnatural transverse, flip

At the level used here $T_{01}^{(-)} = U_{01}^{(-)} = D/2$, real and equal, which is the forward-kinematics relation; the remaining four carry the fitted moduli and phases.

The photon couples to quark q with charge e_q , and the meson wavefunction selects which combination survives. With $\pi^0 = (u\bar{u} - d\bar{d})/\sqrt{2}$ and $\eta_8 = (u\bar{u} + d\bar{d} - 2s\bar{s})/\sqrt{6}$, the coefficient of each convolution is $\sum_q (\text{meson amplitude}) \times e_q$. The strange term drops out, for the reason given below. Writing $X_q \equiv |\langle X \rangle^q|$, this gives

$$\langle X \rangle_{\pi^0 p} = \frac{2X_u + X_d e^{i\varphi_X}}{3\sqrt{2}}, \quad \langle X \rangle_{\eta p} = \frac{2X_u - X_d e^{i\varphi_X}}{3\sqrt{6}k}, \quad \langle X \rangle_{\pi^0 n} = \frac{X_u + 2X_d e^{i\varphi_X}}{3\sqrt{2}}. \quad (5)$$

A superscript on $\langle X \rangle$ is a flavour, a subscript is a channel: $\langle X \rangle^u$ and $\langle X \rangle^d$ are the convolutions of (1), while $\langle X \rangle_{\text{ch}}$ is the channel combination built from them. The relative sign differs between π^0 and η because $e_d < 0$ meets $-d\bar{d}$ in the π^0 and $+d\bar{d}$ in the η . The neutron entry is not new physics: it is the same meson factor with isospin applied to the *target*, $X_u^n = X_d^p$ and $X_d^n = X_u^p$.

Why there is no strange term, and no gluon term. Pseudoscalar meson production filters the C -odd, valence-like combination of quark GPDs – the hard kernel enters antisymmetrised in $x \rightarrow -x$, so quark and antiquark contributions appear with opposite sign. A symmetric sea, $q_{\text{sea}} = \bar{q}_{\text{sea}}$, therefore cancels identically in that combination. Strangeness in the nucleon is pure sea, so $\langle X \rangle^s = 0$ to the extent that $s = \bar{s}$; the approximation fails only through the strangeness asymmetry $s - \bar{s}$, which is small. This is why the $-2s\bar{s}$ piece of η_8 can be dropped in (5) without an additional parameter.

The same C -parity argument removes the gluon contribution altogether, which is what makes π^0 and η production a comparatively clean probe of the valence sector: unlike vector mesons, there is no gluon GPD entering at this order. The singlet component of the η survives only through the η - η' mixing of the decay constants, which is carried by k_L in (7), not by an independent amplitude.

The combination in (5) is a sum of *amplitudes*; the squaring happens only later, in (9). The phase φ_X is therefore physical and not removable: at $\varphi_X = 0$ the η amplitude can pass through an exact zero when $X_d = 2X_u$, whereas for $\varphi_X \neq 0$

$$|2X_u - X_d e^{i\varphi_X}|^2 = 4X_u^2 + X_d^2 - 4X_u X_d \cos \varphi_X \quad (6)$$

has a floor of $4X_u^2 \sin^2 \varphi_X$. Note also that this expression is *even* in φ_X : the unpolarised structure functions cannot determine the sign of the phase, which is fixed only by the interference terms σ_{LT} and $\sigma_{LT'}$.

The factor k is not flavour algebra but meson normalisation, and it enters as a divisor, so $1/k$ is an enhancement of the η channel. In the longitudinal twist-2 sector the meson enters through its decay constant; in the two-mixing-angle scheme ($\theta_8 = -21.2^\circ$, $\theta_1 = -9.2^\circ$, $f_8 = 1.26f_\pi$, $f_1 = 1.17f_\pi$),

$$k_L^{-1} = \left(\cos \theta_8 - \sqrt{2} \frac{f_1}{f_8} \sin \theta_1 \right) \frac{f_8}{f_\pi} = 1.439 \quad \implies \quad k_L = 0.695. \quad (7)$$

In the transverse twist-3 sector the meson enters through the chiral-odd twist-3 distribution amplitude, normalised by $\mu_M \sim m_M^2/(m_u + m_d)$, giving $k_T = 0.863$.

The near cancellation $2X_u - X_d$ in the η channel is what makes that channel sensitive to the flavour separation, and it is the reason η data carry weight far beyond their number.

The amplitudes themselves. Writing $\xi = \frac{x_B}{2-x_B} (1 + M_p^2/Q^2)$, $t' = t - t_{\min}$ and $A = A(Q^2, x_B)$ for the phase-space factor,

$$D = \sqrt{2A(1-\xi^2)} \langle H_T \rangle_{\text{ch}}, \quad U_{01}^{(+)} = \sqrt{A\kappa} \langle \bar{E}_T \rangle_{\text{ch}}, \quad |T_{00}^{(-)}| = \sqrt{A\kappa} \langle L \rangle_{\text{ch}}, \quad (8)$$

with $\kappa = -t'/(8M_p^2)$ and the subscript “ch” the channel combination of (5). Four further parameters unlock what would otherwise be exact relations: ρ_{CE} and ϕ_{CE} give $T_{01}^{(+)}$ a modulus and phase relative to $U_{01}^{(+)}$ (breaking $C = E$), ϕ_w is the phase of $U_{01}^{(+)}$, and ρ_{nf}, ϕ_{nf} give the longitudinal non-flip amplitude $T_{00}^{(+)}$, which vanishes at leading order. The phase of $T_{00}^{(-)}$ is $\delta_0 + \delta_1 t$, common to both flavours – note that this one phase is allowed a t -dependence, unlike φ_X .

1.3 What is free and what is tied

The flavours are not treated symmetrically, and the pattern is a model assumption worth stating. For fit `C1_with_compass`:

block	slope u [GeV ⁻² , at $x_B = 0.25$]	slope d	d/u ratio	phase φ [rad]
$\langle H_T \rangle$	1.491	1.491 (tied to u)	-0.296	-1.261
$\langle \bar{E}_T \rangle$	1.110	5.720 (free)	0.935	0.230
$\langle L \rangle$	1.780	= u	-0.584	0.512

Only $\langle \bar{E}_T \rangle$ is allowed an independent t -slope per flavour, and it uses that freedom: the d slope comes out five times the u slope. For $\langle H_T \rangle$ and for the longitudinal block the two flavours share one slope and differ only in normalisation and phase. Whether $\langle H_T \rangle$ would also want a split slope is not tested here.

1.4 From amplitudes to structure functions

The nine structure functions follow from the Fourier dictionary, verified in the note against the QED photon density matrix:

$$\begin{aligned} \sigma_L &= \sum |T_{00}|^2, & \sigma_T &= \sum |T_{01}|^2 + \sum |U_{01}|^2, & \sigma_{TT} &= \sum |T_{01}|^2 - \sum |U_{01}|^2, \\ \sigma_{LT} + i\sigma_{LT'} &= -\sqrt{2} \sum T_{00}^* T_{01}, & \sigma_{LT'}^{LL} + i\sigma_{LT}^{UL} &= -\sqrt{2} \sum T_{00}^* U_{01}, & \sigma_{T'} + i\sigma_{TT}^{UL} &= 2 \sum T_{01}^* U_{01}. \end{aligned} \quad (9)$$

Only the moduli of T_{01} and U_{01} enter σ_T and σ_{TT} ; their *relative phase* appears only in $\sigma_{T'}$. This observation matters later: it is why one polarised-target experiment is the sole constraint on that phase.

The measured cross section, for beam helicity $h = \pm 1$ and a *longitudinally* polarised target of polarisation Λ , is

$$\begin{aligned} \frac{d^2\sigma}{dt d\phi} &= \frac{1}{2\pi} \left[\sigma_T + \epsilon \sigma_L + \epsilon \cos 2\phi \sigma_{TT} + \sqrt{2\epsilon(1+\epsilon)} \cos \phi \sigma_{LT} + h \sqrt{2\epsilon(1-\epsilon)} \sin \phi \sigma_{LT'} \right. \\ &\quad + \Lambda \left(\sqrt{2\epsilon(1+\epsilon)} \sin \phi \sigma_{LT}^{UL} + \epsilon \sin 2\phi \sigma_{TT}^{UL} \right) \\ &\quad \left. + h \Lambda \left(\sqrt{1-\epsilon^2} \sigma_{T'} + \sqrt{2\epsilon(1-\epsilon)} \cos \phi \sigma_{LT'}^{LL} \right) \right], \end{aligned} \quad (10)$$

with ϕ in the Trento convention. The first line, which is all that survives at $\Lambda = 0$, is identical to eq. (B4) of the CLAS6 paper and eq. (2) of COMPASS. The second line is the target-spin asymmetry A_{UL} and the third the double-spin asymmetry A_{LL} ; dividing each term by $\sigma_0 = \sigma_T + \epsilon \sigma_L$ gives the four moments that are measured and that enter this fit,

$$A_{UL}^{\sin \phi} = \frac{\sqrt{2\epsilon(1+\epsilon)} \sigma_{LT}^{UL}}{\sigma_0}, \quad A_{UL}^{\sin 2\phi} = \frac{\epsilon \sigma_{TT}^{UL}}{\sigma_0}, \quad A_{LL}^{\text{const}} = \frac{\sqrt{1-\epsilon^2} \sigma_{T'}}{\sigma_0}, \quad A_{LL}^{\cos \phi} = \frac{\sqrt{2\epsilon(1-\epsilon)} \sigma_{LT'}^{LL}}{\sigma_0}. \quad (11)$$

Two remarks on the scope of eq. (10). A *transversely* polarised target requires structure functions this model does not carry, so only the longitudinal case is written. And the hadronic tensor from which the first line is usually derived — eq. (10) of the radiative-correction note — is summed over the target spin and is built only from the photon basis vectors X, Y, ℓ ; the last two lines require the terms linear in the target spin vector S^μ , which that tensor does not have. The generator inherits the same restriction: it samples the first line only.

2 The data

Thirteen data sets are loaded from a single workbook, `pi0_eta_database.xlsx`. Nothing enters the fit from any other source: a second source is how an old, mis-weighted version of one set once survived unnoticed. Table 1 lists everything, fitted or not.

2.1 Provenance of the uncertainties

Every weight in the fit now comes from a publication. Three defects were repaired on 5 September 2026 and are recorded here because they changed the answer, not merely the bookkeeping.

Table 1: The database and the fit. A set marked $--$ is not fitted; its χ^2 is then a blind prediction. CLAS12 cross sections are preliminary; the published De Masi α are the same measurement as the ϕ distributions and would be double counting.

set	key	fitted	points	χ^2	per point
CLAS6 π^0 p	clas6_pi0	yes	288	482.4	1.68
CLAS6 η p	clas6_eta	yes	207	217.0	1.05
Hall A 2017, π^0 n	halla_n	yes	12	20.7	1.72
Hall A 2016, π^0 p	halla_y16	yes	21	95.3	4.54
Hall A 2011, π^0 p	halla_y11	yes	112	143.3	1.28
Hall A 2021, π^0 p	halla_y21	yes	144	358.2	2.49
CLAS12 cross sections	clas12_xs	—	438	3853.3	8.80
COMPASS 2025	compass	yes	26	65.4	2.51
De Masi, published α	bsa_demasi	—	56	82.3	1.47
Zhao, η BSA	bsa_zhao	yes	12	13.4	1.12
CLAS12 $\sigma_{LT'}/\sigma_0$	bsa_clas12	yes	30	45.6	1.52
De Masi, $A_{LU}(\phi)$	bsa_demasi_phi	yes	663	620.7	0.94
eg1-dvcs, pol. target	eg1	yes	40	71.7	1.79

De Masi carried no systematic at all. The CLAS physics database export gives statistical errors only, so 703 ϕ points were the most heavily weighted set in the sample. Ref. PRC **77** 042201(R), p. 3, quotes 0.016 from event selection and from the choice of fit function; it is now applied to all 703 points and to the 62 published moments.

The Hall A neutron systematics exist only as a picture. PRL **118** 222002 gives no table; the point-to-point uncertainties appear only as the shaded bands of its figure 5. They were read from the vector PDF, with the vertical calibration taken not from the axis but from the published error bars themselves — half a bar in figure units against the quoted statistical error. The calibration closes to 0.0 % in the σ_T panel and 1.8 % in σ_{LT} and σ_{TT} . The bands are asymmetric, in places by a factor three; they are symmetrised as (up + down)/2, and the 3.1 % normalisation they contain is removed in quadrature because it is fitted separately.

An invented 10 % was removed. Hall A 2016 and the neutron had been carrying a flat 10 % of the author’s own making. It was wrong in both directions at once: it concealed a real discrepancy in the 2016 proton set (3.93 $\rightarrow$ 5.78 per point once removed) and inflated a false one in the neutron, whose true bands are larger than 10 %. An invented uncertainty is worse than a missing one, because it looks like knowledge.

2.2 Normalisation uncertainties

Three sets carry an overall normalisation that multiplies every point together: 3.5 % for De Masi (beam polarisation), 3.12 % for Hall A 2016 (ref. PRL **117** 262001, table III) and 3.1 % for the neutron. Adding these in quadrature to the per-point errors would discard the correlation and smear each set instead of letting it move as a whole. They are therefore fitted as one free scale per set,

$$r = \frac{\text{data} - \lambda \text{model}}{\sigma}, \quad \text{with a penalty} \quad \left(\frac{\lambda - 1}{\sigma_{\text{norm}}} \right)^2 \text{ counted in } \chi^2, \quad (12)$$

and the number of degrees of freedom reduced by the number of λ ; otherwise a nuisance would be a free way to improve χ^2/ndf . Table 2 gives the results. Hall A 2016 is the one to notice: its normalisation is dragged nearly two standard deviations and the set still does not fit.

Table 2: Fitted normalisations.

set	prior	λ	pull
bsa_demasi_phi	3.50 %	1.0586 ± 0.0326	$+1.67 \sigma$
halla_n	3.10 %	0.9776 ± 0.0304	-0.72σ
halla_y16	3.12 %	1.0583 ± 0.0253	$+1.87 \sigma$

3 The fit

3.1 Constraints

Two constraints are imposed for stated physical reasons rather than to improve χ^2 .

$n_{Q,d} = n_{Q,u}$ **in both transverse sectors.** A generalised form factor is (GPD) $\times$ (hard subprocess). The GPD carries no Q^2 dependence, and the subprocess is flavour-blind up to the quark charge. Transversity does not mix with gluons, so its evolution is non-singlet and flavour-blind as well. Left free, the fitted $n_{Q,d} - n_{Q,u}$ came out -7.0 and $+2.8$ in successive fits, with flavour ratios running by an order of magnitude across Q^2 .

$b_d = b_u$ **for H_T .** Left free, the fit drives b_d to whatever ceiling it is given — 12, 18, 25 — buying two units of χ^2 while the normalisation N_d compensates by a factor sixteen. It is a flat valley, not a measurement.

In addition every t -slope is required non-negative over $x_B \in [0.05, 1]$, not only where there are data: without it H_T^u came out with $b = -0.26$, so that the form factor *grew* with $|t|$ above $x_B = 0.785$.

3.2 Parameters

Tables 3 and 4 give all 28 fitted parameters with errors from the Jacobian at the minimum. A dagger marks a parameter sitting on a bound, where the quoted error is one-sided and the value is not a measurement.

Table 3: Block parameters, eq. (3). Every block is a *convolution* $\langle X \rangle$, not a GPD: no GPD is fitted anywhere in this work. N in $\text{nb}^{1/2}/\text{GeV}$, b and b' in GeV^{-2} , n_Q dimensionless.

block	N	b	b'	n_Q
$\langle H_T \rangle^u$	20.376 ± 0.608	$0.000 \pm 0.043^\dagger$	-1.076 ± 0.060	1.260 ± 0.032
$\langle H_T \rangle^d$	-6.023 ± 5.464	$= b_u$	$= b'_u$	$= n_{Q,u}$
$\langle \bar{E}_T \rangle^u$	60.867 ± 2.909	0.672 ± 0.051	-0.316 ± 0.037	1.141 ± 0.066
$\langle \bar{E}_T \rangle^d$	226.967 ± 21.682	5.283 ± 0.358	$= b'_u(\bar{E}_T)$	$= n_{Q,u}(\bar{E}_T)$
$\langle L \rangle$	35.796 ± 4.449	$0.000 \pm 0.043^\dagger$	-1.284 ± 0.103	0.394 ± 0.144

Two features deserve comment. N_d of H_T is consistent with zero: the d -quark transversity normalisation is not determined by these data at all. And the d/u phase of $\bar{E}_T$ is likewise consistent with zero, while only the H_T phase differs from zero, by about 4.5σ .

3.3 Derived quantities

At $Q^2 = 1.75 \text{ GeV}^2$, $x_B = 0.36$, $-t = 0.27 \text{ GeV}^2$:

$$\left. \frac{d}{u} \right|_{H_T} = -0.296, \quad \left. \frac{d}{u} \right|_{\bar{E}_T} = 0.935, \quad \left. \frac{n}{p} \right|_{\sigma_{TT}} = 1.049, \quad \frac{\sigma_L}{\sigma_T} = 0.080.$$

The n/p ratio carries a caveat that is not a formality; see section 5.

Table 4: Remaining parameters, with the interval each was allowed.

parameter	value	lower	upper
$R_L = N_d/N_u \langle \langle L \rangle \rangle$	-0.670 ± 0.180	-2.00	2.00
δ_0	1.736 ± 0.083	-3.14	3.14
δ_1	0.692 ± 0.089	-3.00	3.00
ρ_{CE}	0.169 ± 0.021	0.00	1.50
ϕ_{CE} (frozen)	0.000 ± 0.000	-3.14	3.14
ϕ_w	-0.629 ± 0.126	-3.14	3.14
ρ_{nf}	1.787 ± 0.209	0.00	3.00
ϕ_{nf}	2.061 ± 0.149	-3.14	3.14
$\varphi_{d/u}(\langle \langle H_T \rangle \rangle)$	-1.261 ± 0.292	-3.14	3.14
$\varphi_{d/u}(\langle \langle \bar{E}_T \rangle \rangle)$	0.230 ± 0.429	-3.14	3.14
$\varphi_{d/u}(\langle \langle L \rangle \rangle)$	0.512 ± 0.325	-3.14	3.14

4 What the leave-one-out analysis shows

Each data set was removed in turn and the fit repeated, in two series: without and with Hall A 2021. A set's χ^2 when it is *not* fitted is a blind prediction, and is the useful number: a set that is left out and still described brings no unique information.

Table 5: Blind χ^2 per point of the set that was removed, and the effect on two derived quantities. Series 8 has 9 sets and no Hall A 2021; series D has 10, with it, and is fitted to the corrected x_B .

removed	blind, series 8	blind, series D	$d/u _{H_T}$	n/p
— (reference)	—	—	-0.314	1.070
CLAS6 π^0	2.89	3.69	-0.328	0.974
CLAS6 η	218.9	2.90	-0.463	1.236
Hall A neutron	10.06	13.48	+0.407	2.195
Hall A 2016	6.18	6.03	-0.340	1.046
Hall A 2011	2.66	1.81	-0.348	1.098
Hall A 2021	—	11.05	-0.238	1.037
De Masi ϕ	1.02	1.02	-0.283	1.080
Zhao	2.25	1.73	-0.247	1.093
CLAS12 BSA	2.88	2.59	-0.335	1.055
egl-dvcs	40.16	66.54	-0.403	0.652

Hall A 2021 has taken the flavour separation over from η . Without it, removing the η data left the model predicting η at 219 per point. With it in the fit that collapses to 2.90. Hall A 2021 reaches $x_B = 0.60$, the highest in the database, and now covers the region that only the η cancellation used to probe. η still moves $d/u|_{H_T}$ from -0.314 to -0.463 when dropped; it is no longer irreplaceable.

The reference fit is reproducible. Removing Hall A 2021 from series D reproduces the series 8 reference to four figures, from a different starting point. The minima are real and not wherever the optimiser happened to land.

5 What the data determine weakly, or not at all

This section is the point of the exercise, and is stated as plainly as the numbers allow.

n/p is soft against the model, not only against the data. The neutron channel is $\langle X \rangle \propto X_u + 2X_d$ against the proton's $2X_u + X_d$, so the d block enters the neutron with twice the weight it

has in the proton. Since $\sigma_{TT} = \sum |T_{01}|^2 - \sum |U_{01}|^2$ and U_{01} is built from $\bar{E}_T$, the ratio n/p in σ_{TT} is essentially a statement about $\bar{E}_T^d$ relative to $\bar{E}_T^u$.

The fitted t -slopes of those two blocks differ by a factor eight, $b_d = 5.28 \pm 0.36$ against $b_u = 0.67 \pm 0.05$. That invites the suspicion that it is the flat valley which forced $b_d = b_u$ in the H_T sector, where the same tie costs two units of χ^2 . It is not: tying them in $\bar{E}_T$ costs 194.5 units for one parameter, which excludes the tie decisively.

But look at who pays. The cost falls on the *proton* sets — CLAS6 π^0 +73.8, Hall A 2016 +50.3, eg1 +34.7, Hall A 2011 +26.8 — while the neutron, the set whose four points are supposed to be carrying this ratio, fits *better* under the tie, $1.72 \rightarrow 0.65$ per point, and CLAS6 η improves as well. Under that constrained variant n/p is 0.52 instead of 1.05: half the value, at a χ^2 the neutron itself prefers.

So the steep d slope is a requirement of the proton data, and the neutron points do not pin n/p — they are comfortable with a value twice smaller. The quoted 1.049 is what the unconstrained fit returns, and the 194 units say the data prefer it, but the number is soft against a change of model assumption in a way that no error bar expresses. Its history over one day of minor and justified changes — corrected systematics, a normalisation nuisance, a repaired bound, a corrected x_B — runs 1.00, 1.02, 0.98, 1.04, 1.06, 1.05, with χ^2/ndf moving in the third decimal. A fifth neutron point would be worth more than a thousand proton points, and a measurement that constrained $\bar{E}_T^d$ directly would be worth more still.

One experiment fixes a phase. $\sigma_{T'} = 2 \text{Re}\langle T_{01}, U_{01} \rangle$ by eq. (9); the unpolarised data fix only the moduli. The A_{LL} moments of eg1-dvcs are the only observable in the database that sees this interference, and when eg1 is left out its own blind χ^2 is 41 per point, with ϕ_w swinging from -0.70 to $+2.90$ — essentially a sign flip of $\sigma_{T'}$. Forty points from one experiment.

The experiments do not agree on the sign of σ_{LT} . This was at first read as a failure of the model, because the blind prediction of the CLAS12 cross sections gives 20.4 per point in σ_{LT} against 2.4 in σ_U and 3.6 in σ_{TT} . It is not. Looking at the data rather than at the χ^2 :

set	$\langle \sigma_{LT} \rangle$	fraction negative
Hall A 2011	−77.6	0.86
Hall A 2016	−42.6	0.86
Hall A 2021	−7.0	0.83
CLAS6 π^0	+0.07	0.43
CLAS6 η	−3.2	0.58
CLAS12 (preliminary)	+8.3	0.24

All three Hall A measurements are firmly negative, CLAS6 sits near zero, and CLAS12 is positive. The model follows the sets that are in the fit and comes out negative at CLAS12 kinematics, which is where the 20.4 comes from. The pull grows monotonically with x_B , from +2.1 below 0.2 to +5.2 above 0.4 — a pattern that looks more like a convention than like physics, and the CLAS12 numbers are preliminary. In every set that is fitted, σ_{LT} is described as well as everything else: 1.99 for CLAS6 π^0 , 0.67 for CLAS6 η , 1.64 for Hall A 2011, 2.55 for Hall A 2021. Hall A 2016 gives 4.85, but its σ_T gives 4.87 — that set is poorly described as a whole, not in this one quantity.

Nothing here is settled. Either CLAS12 will change sign when it is finalised, or one of the two families has a convention error, or the σ_{LT} of the proton really does change sign somewhere between these kinematics. It matters for anything built on the model: σ_{LT} carries the whole $\cos \phi$ modulation of the cross section.

A single power of Q^2 does not span the range. Hall A 2016 and Hall A 2021 disagree with the model in opposite directions at the two ends: at $Q^2 = 1.5$ the model undershoots σ_T

(1616 ± 145 measured against 1135), at $Q^2 = 8.3$ it overshoots σ_U by a factor 1.8. Hall A 2016 does not fit at all — 4.5 per point with its normalisation already dragged 1.8σ — and these are the only Rosenbluth-separated proton points that exist. Admitting Hall A 2021 to the fit brings it from 13.1 to 2.34 per point, but the asymmetries pay: eg1 from 1.36 to 1.81, the CLAS12 beam-spin asymmetry from 1.04 to 1.53. The compromise is real and it is visible.

6 Comparison with Goloskokov–Kroll

Two independent comparisons put the GK prediction for $\sigma_{LT'}$ far below the data. The published GK curve for the η beam-spin asymmetry (Zhao *et al.*, PLB **789** 426, figure 5) is near zero across all four bins where the data are 0.1 to 0.3. For π^0 , a GK calculation over the 60 De Masi bins gives $\alpha \approx 0.01$ – 0.025 against a published median of 0.080. The present model gives a median 0.077 on the same bins, but that is a fit, not a prediction; the fair statement is that when the De Masi ϕ distributions are removed from the fit entirely they are still predicted at 1.02 per point, so the magnitude of $\sigma_{LT'}$ demanded by all the *other* data is about five times what GK produces.

Throughout, α is the amplitude of $A_{LU} = \alpha \sin \phi / (1 + \beta \cos \phi + \gamma \cos 2\phi)$ with A_{LU} already divided by the beam polarisation, so $\alpha = \sqrt{2\epsilon(1-\epsilon)} \sigma_{LT'} / (\sigma_T + \epsilon \sigma_L)$ and not the bare ratio.

7 Figures

Every figure shows the data of one set against the fit. A title reading NOT IN THE FIT marks a blind prediction. Data are drawn in red, the model in blue. Each panel carries its own kinematics and χ^2 ; the model curve includes the fitted normalisation of table 2 where the set has one, so the curve is what the fit actually compares against the points.

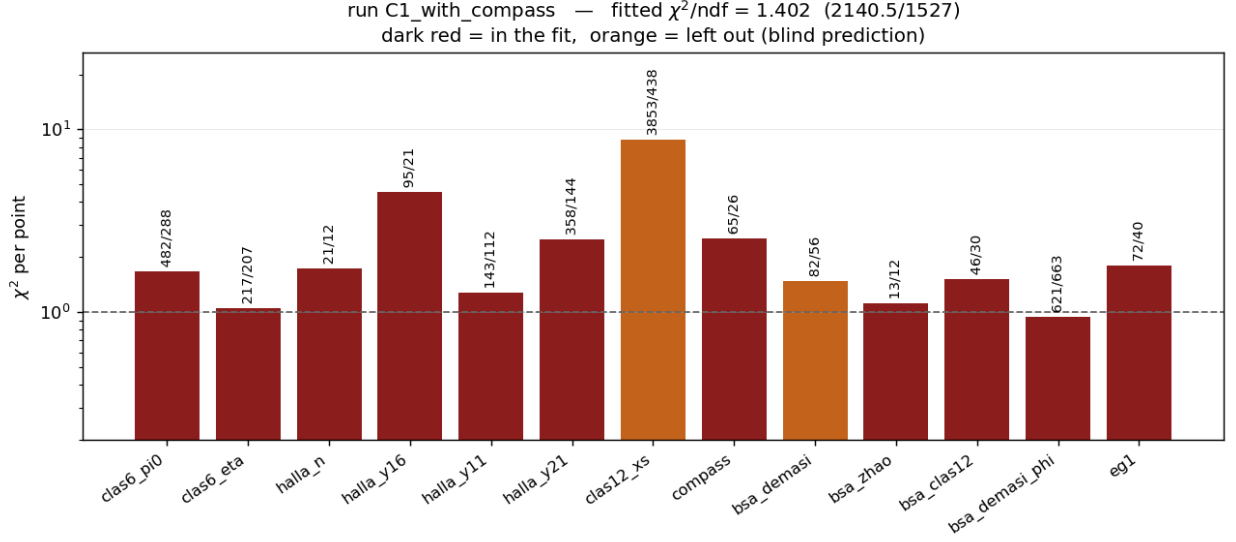


Figure 1: All thirteen sets at a glance: χ^2 per point, fitted and blind.

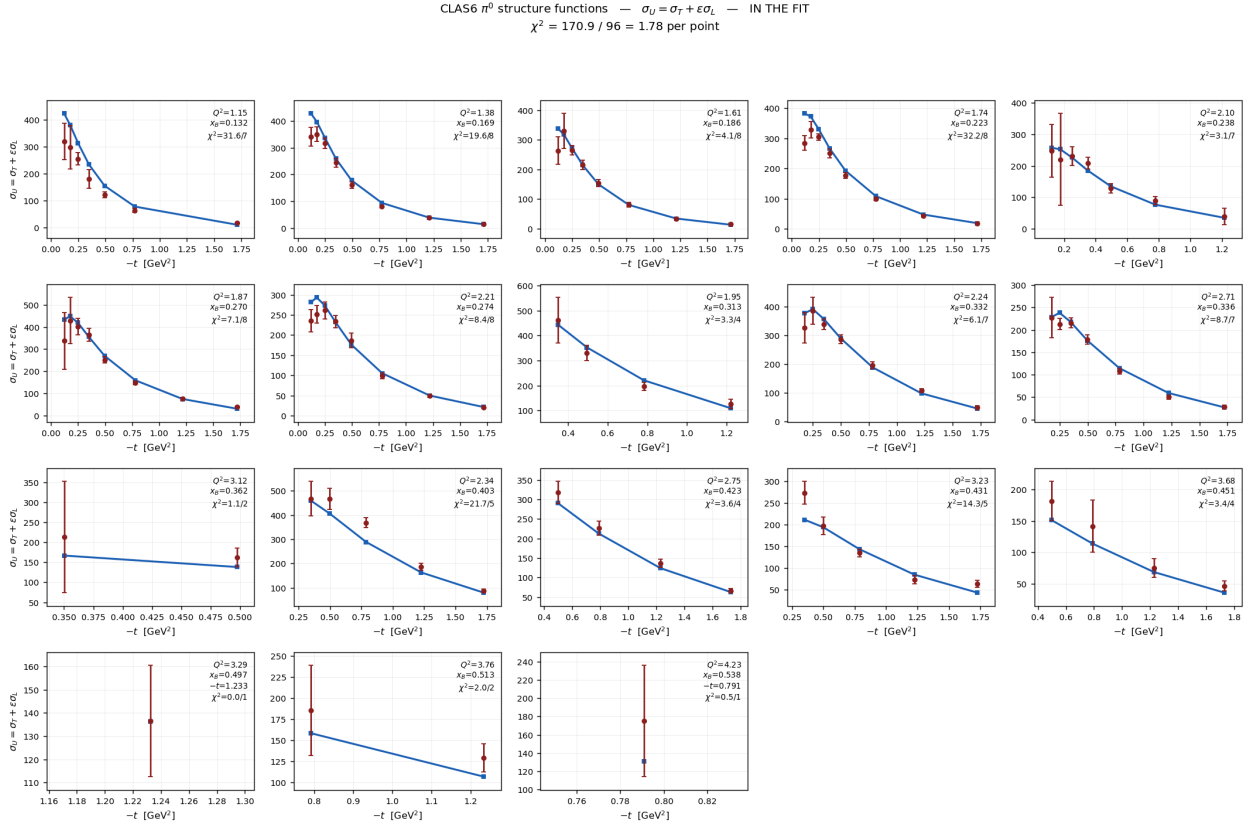


Figure 2: CLAS6 π^0 , $\sigma_U = \sigma_T + \epsilon\sigma_L$, with re-computed radiative corrections.

CLAS6 π^0 structure functions — σ_{LT} — IN THE FIT
 $\chi^2 = 191.3 / 96 = 1.99$ per point

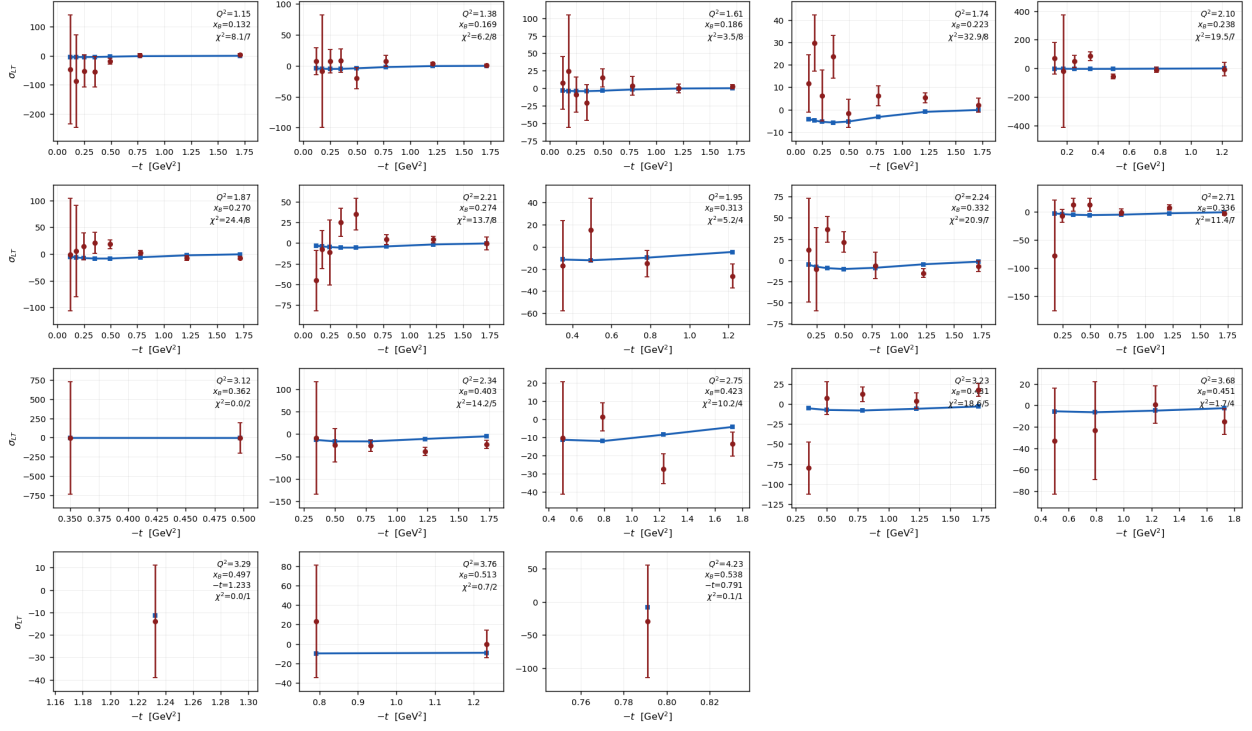


Figure 3: CLAS6 π^0 , σ_{LT} .

CLAS6 π^0 structure functions — σ_{TT} — IN THE FIT
 $\chi^2 = 120.2 / 96 = 1.25$ per point

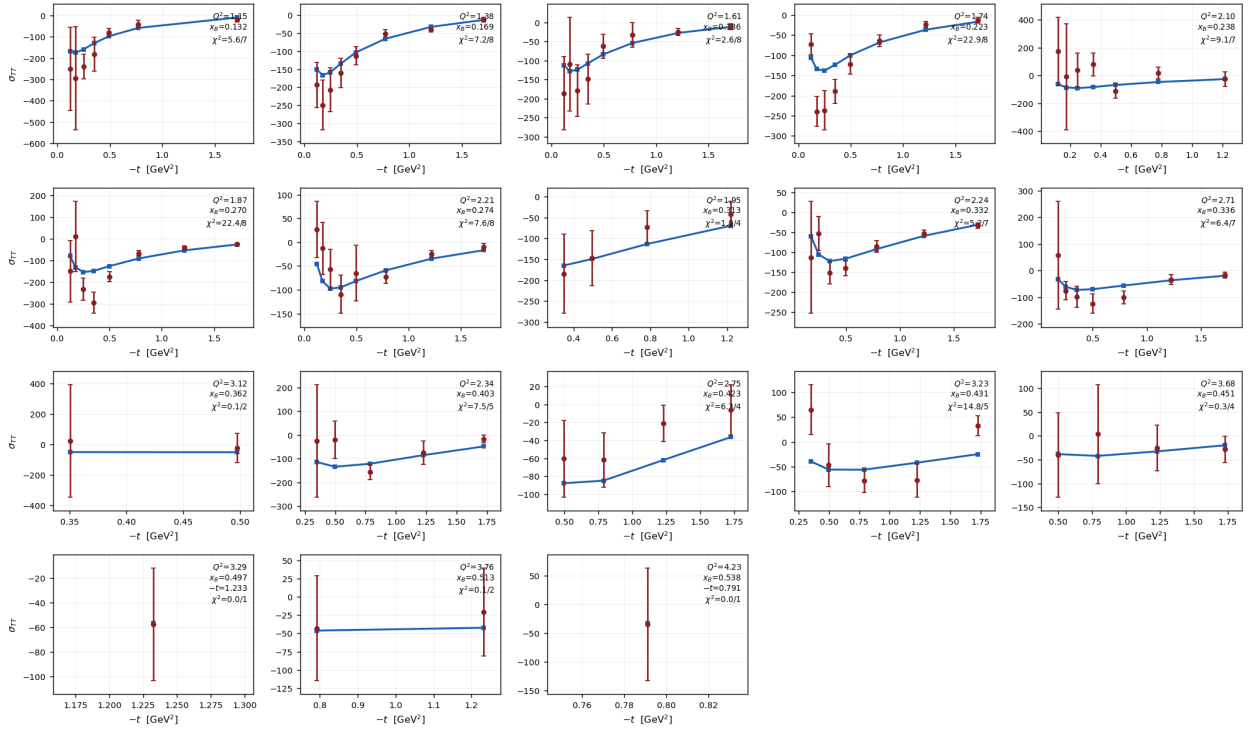


Figure 4: CLAS6 π^0 , σ_{TT} .

CLAS6 η structure functions — $\sigma_U = \sigma_T + \epsilon\sigma_L$ — IN THE FIT
 $\chi^2 = 100.9 / 69 = 1.46$ per point

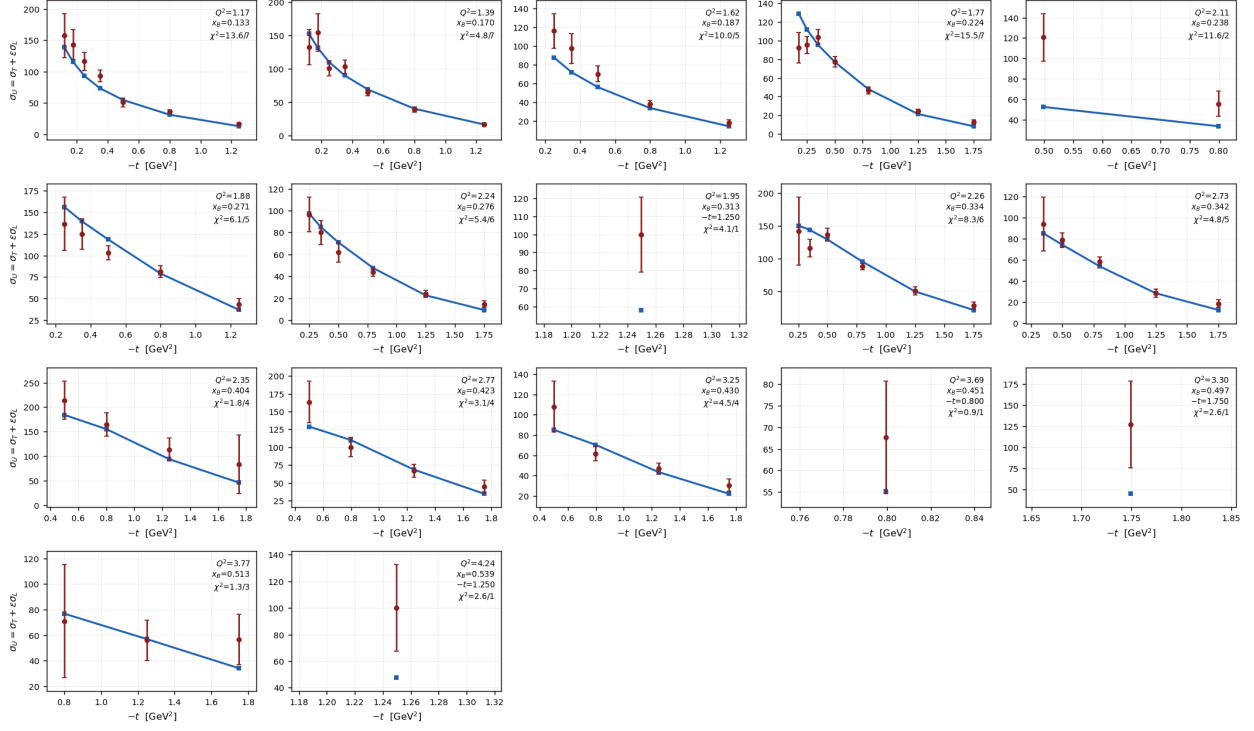


Figure 5: CLAS6 η , σ_U . This channel probes $2X_u - X_d$.

CLAS6 η structure functions — σ_{LT} — IN THE FIT
 $\chi^2 = 46.3 / 69 = 0.67$ per point

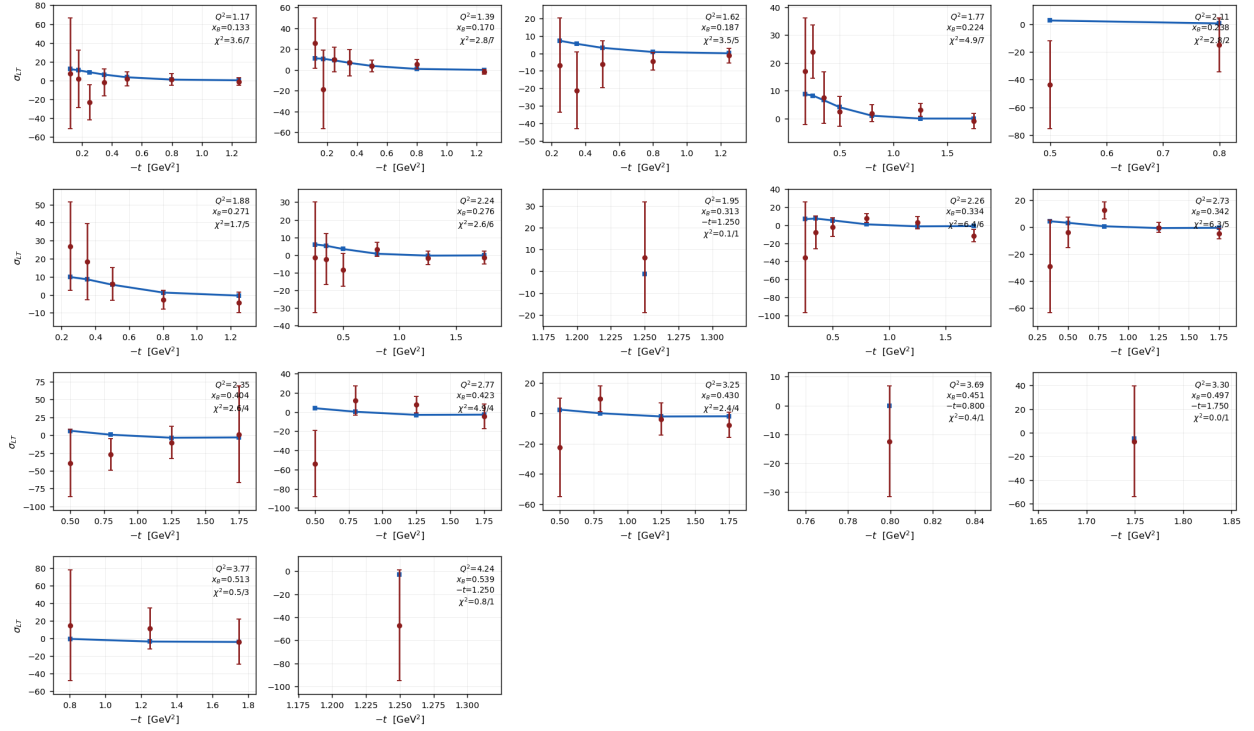


Figure 6: CLAS6 η , σ_{LT} .

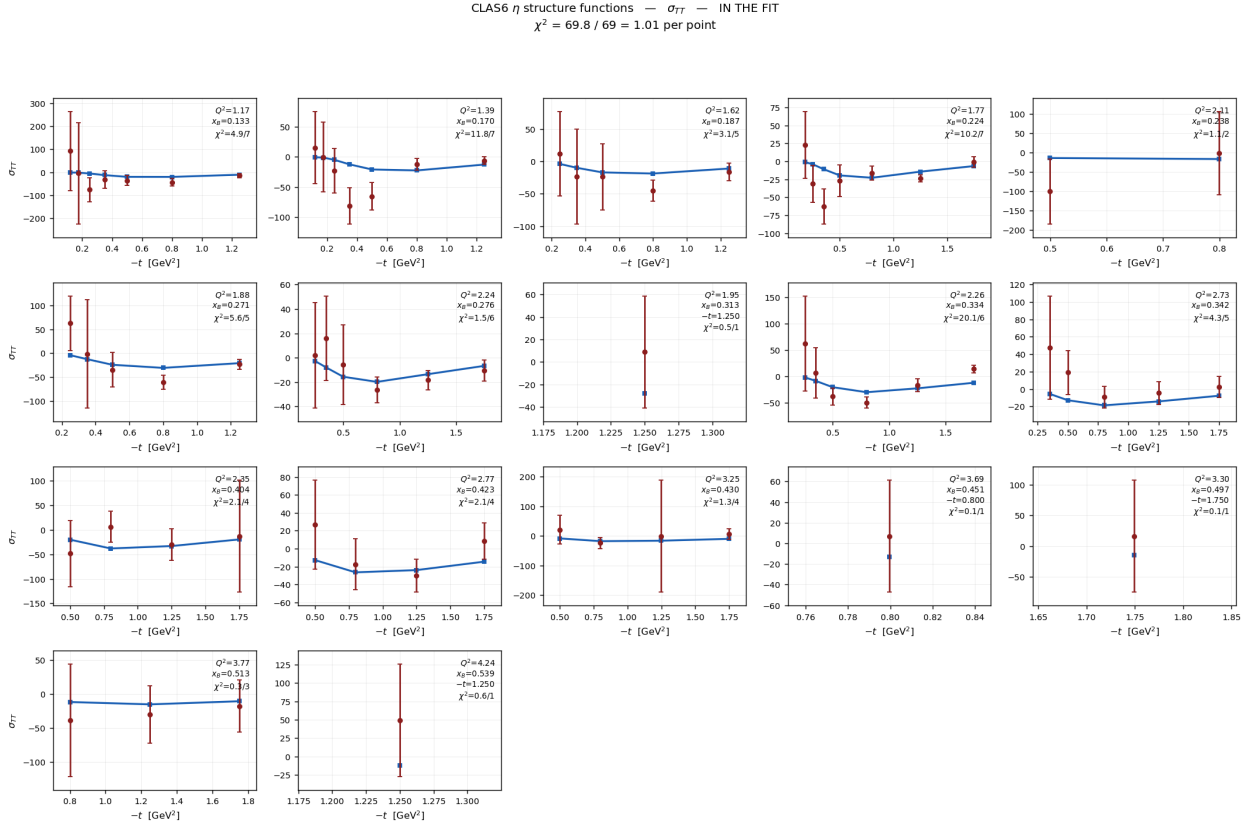


Figure 7: CLAS6 η , σ_{TT} .

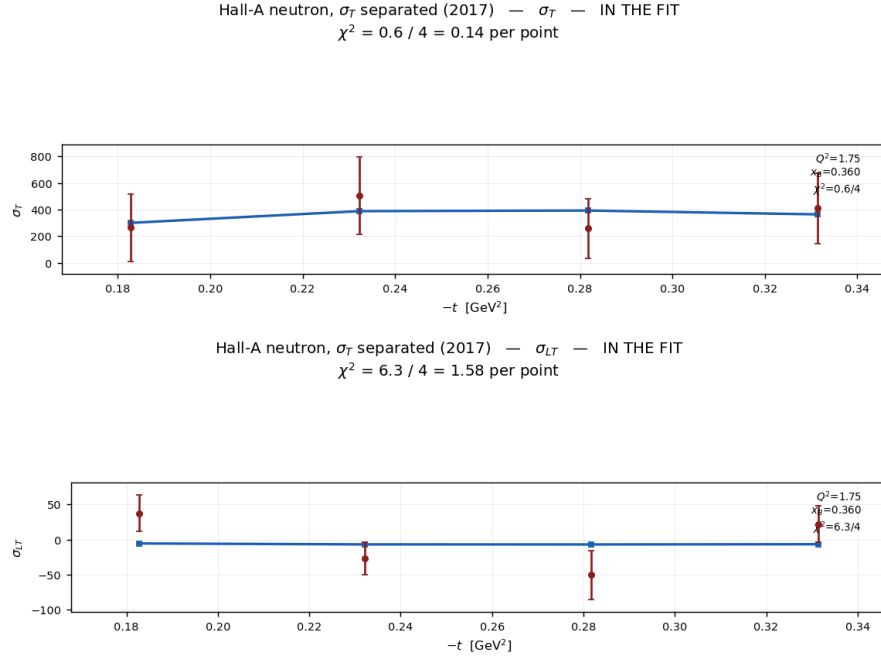


Figure 8: Hall A 2017, π^0 off the *neutron*: σ_T and σ_{LT} . Four points, uncertainties of 42 to 90 %, and the sole constraint on n/p .

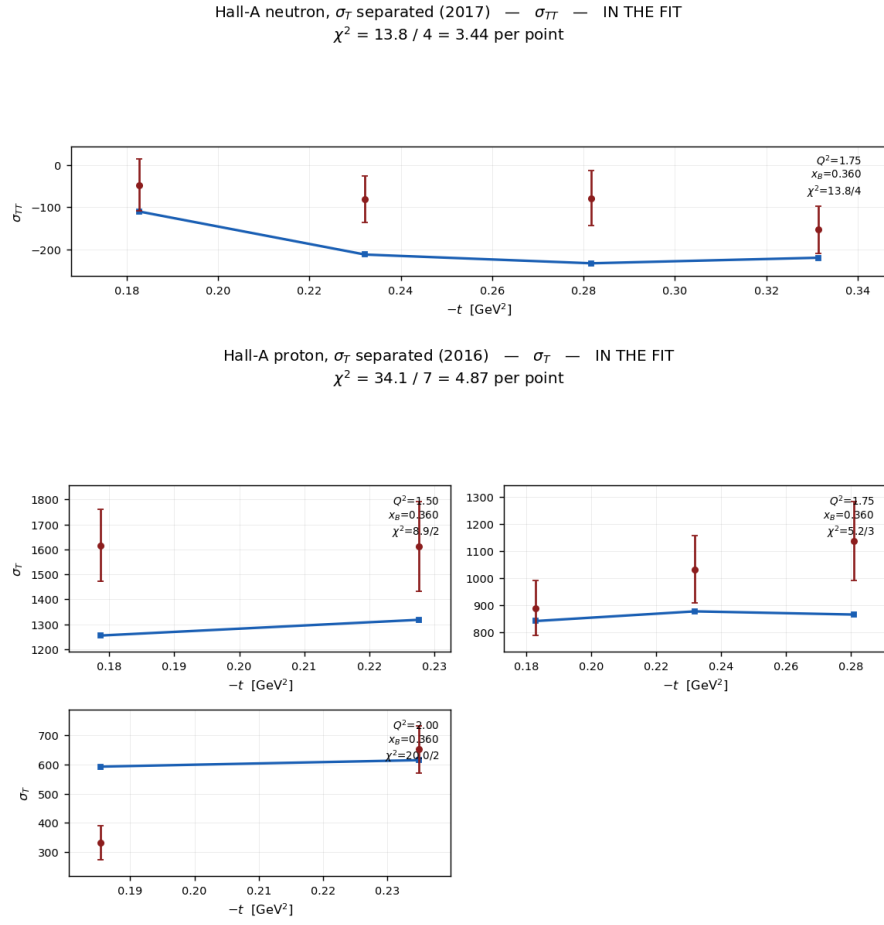
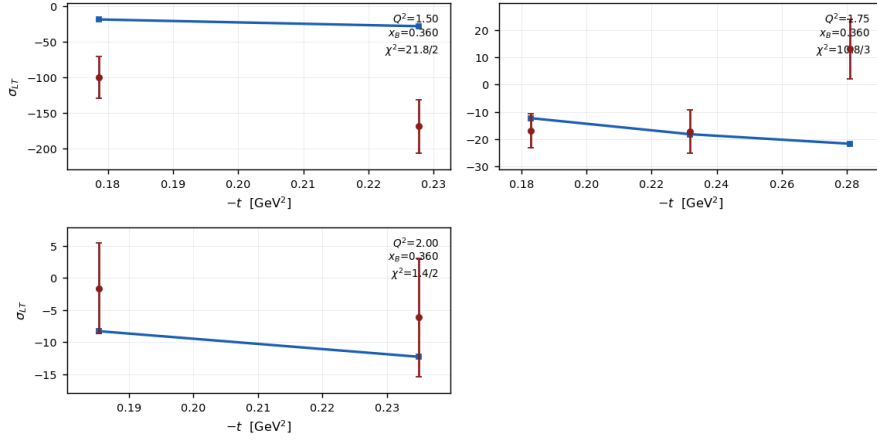


Figure 9: Neutron σ_{TT} (top) and Hall A 2016 proton σ_T (bottom). The latter is the worst-described set in the fit at 4.5 per point, with its normalisation already pulled 1.8σ .

Hall-A proton, σ_T separated (2016) — σ_{LT} — IN THE FIT
 $\chi^2 = 33.9 / 7 = 4.85$ per point



Hall-A proton, σ_T separated (2016) — σ_{TT} — IN THE FIT
 $\chi^2 = 27.3 / 7 = 3.91$ per point

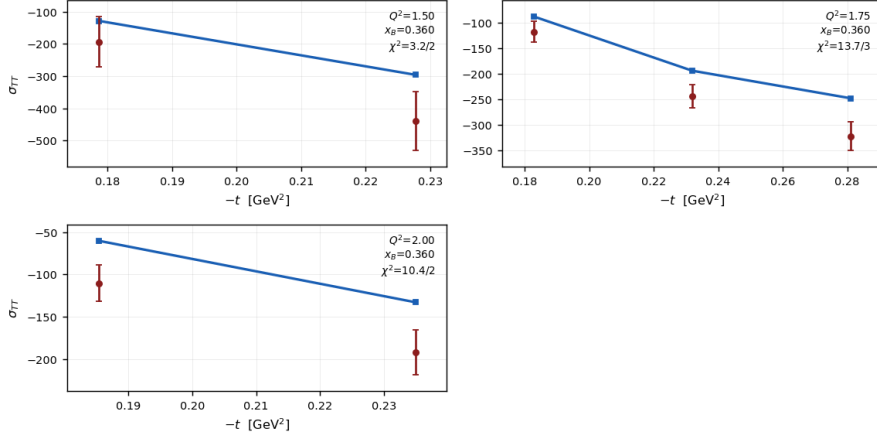


Figure 10: Hall A 2016, σ_{LT} and σ_{TT} .

Hall-A proton 6 GeV (2011) — $\sigma_U = \sigma_T + \varepsilon\sigma_L$ — IN THE FIT
 $\chi^2 = 42.0 / 28 = 1.50$ per point

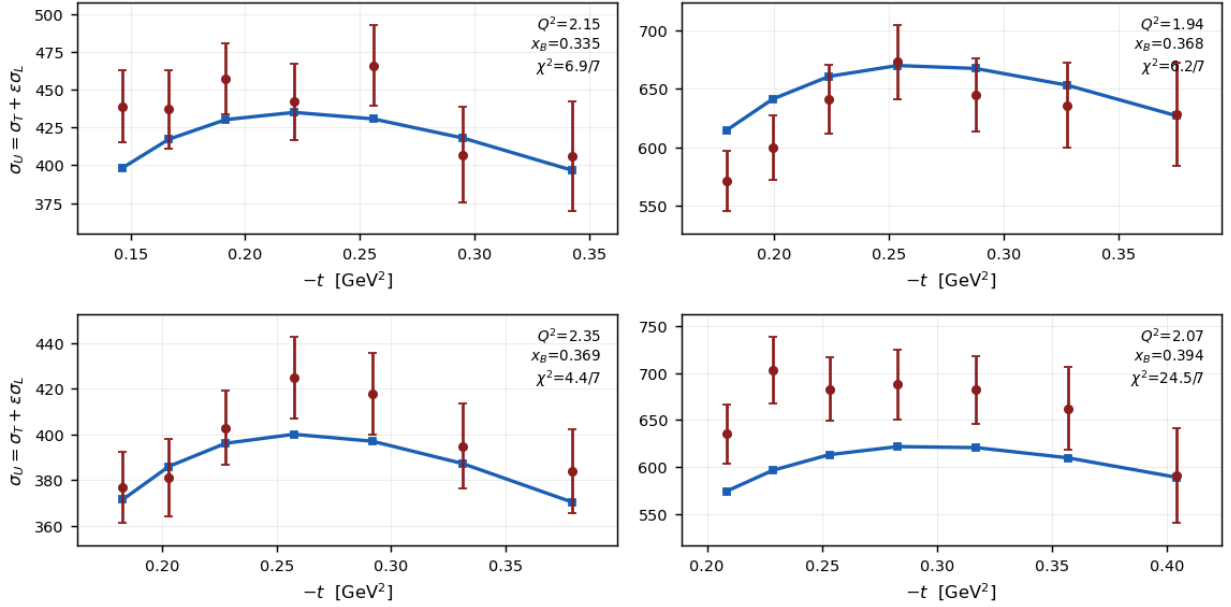
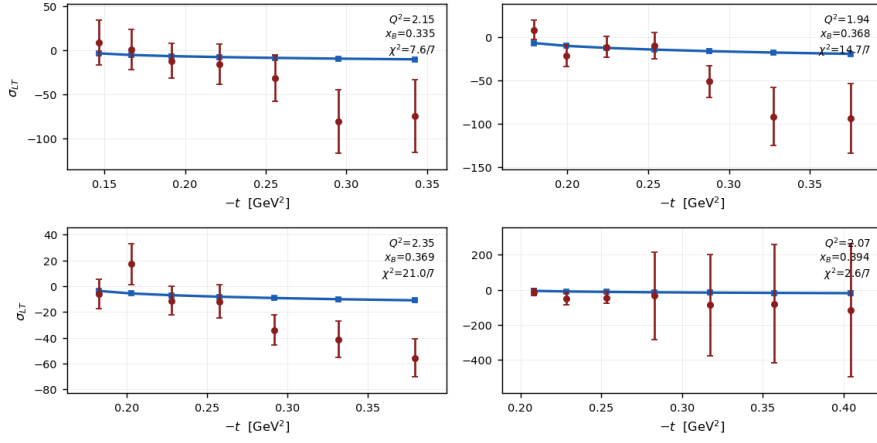


Figure 11: Hall A 2011, σ_U . This set is published in the Drechsel–Tiator normalisation; σ_{LT} and $\sigma_{LT'}$ are converted by $\gamma = 2Mx_B/\sqrt{Q^2}$ before comparison.

Hall-A proton 6 GeV (2011) — σ_{LT} — IN THE FIT
 $\chi^2 = 45.9 / 28 = 1.64$ per point



Hall-A proton 6 GeV (2011) — σ_{TT} — IN THE FIT
 $\chi^2 = 39.4 / 28 = 1.41$ per point

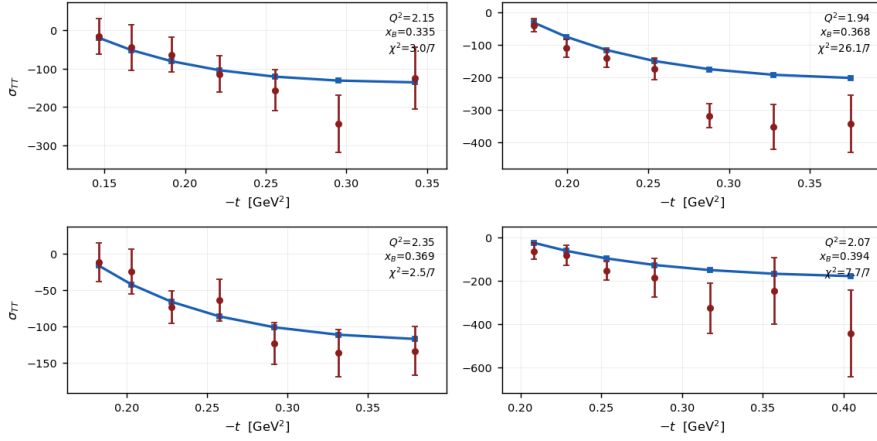


Figure 12: Hall A 2011, σ_{LT} and σ_{TT} .

Hall-A proton 6 GeV (2011) — $\sigma_{LT'}$ — IN THE FIT
 $\chi^2 = 16.0 / 28 = 0.57$ per point

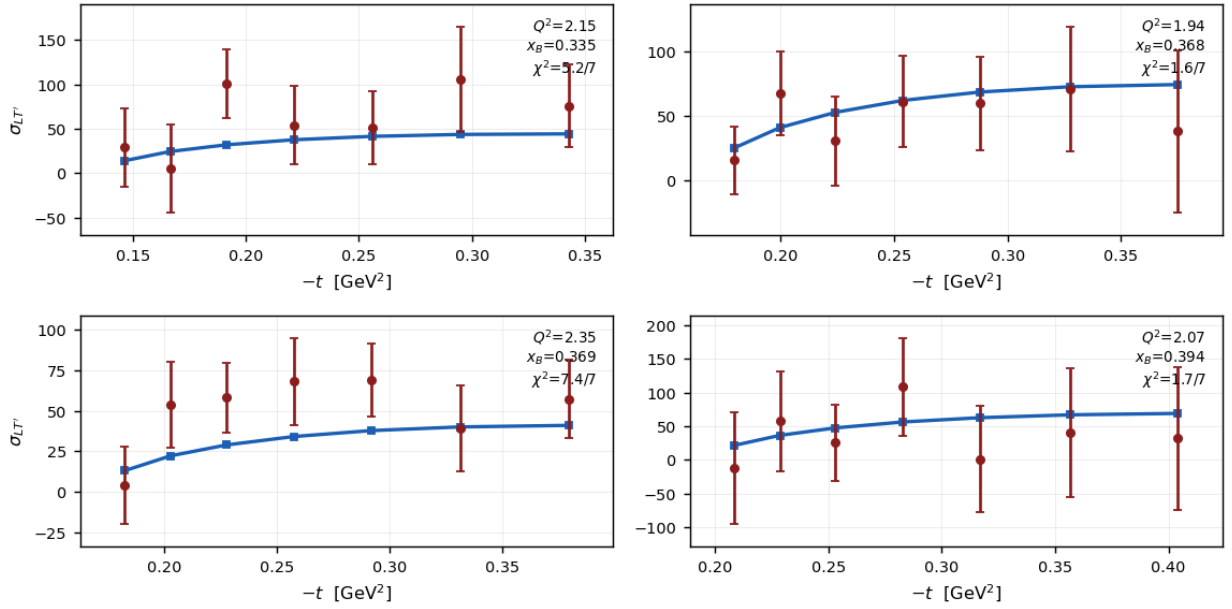


Figure 13: Hall A 2011, $\sigma_{LT'}$.

Hall-A proton 12 GeV (2021) — $\sigma_U = \sigma_T + \varepsilon \sigma_L$ — IN THE FIT
 $\chi^2 = 114.0 / 36 = 3.17$ per point

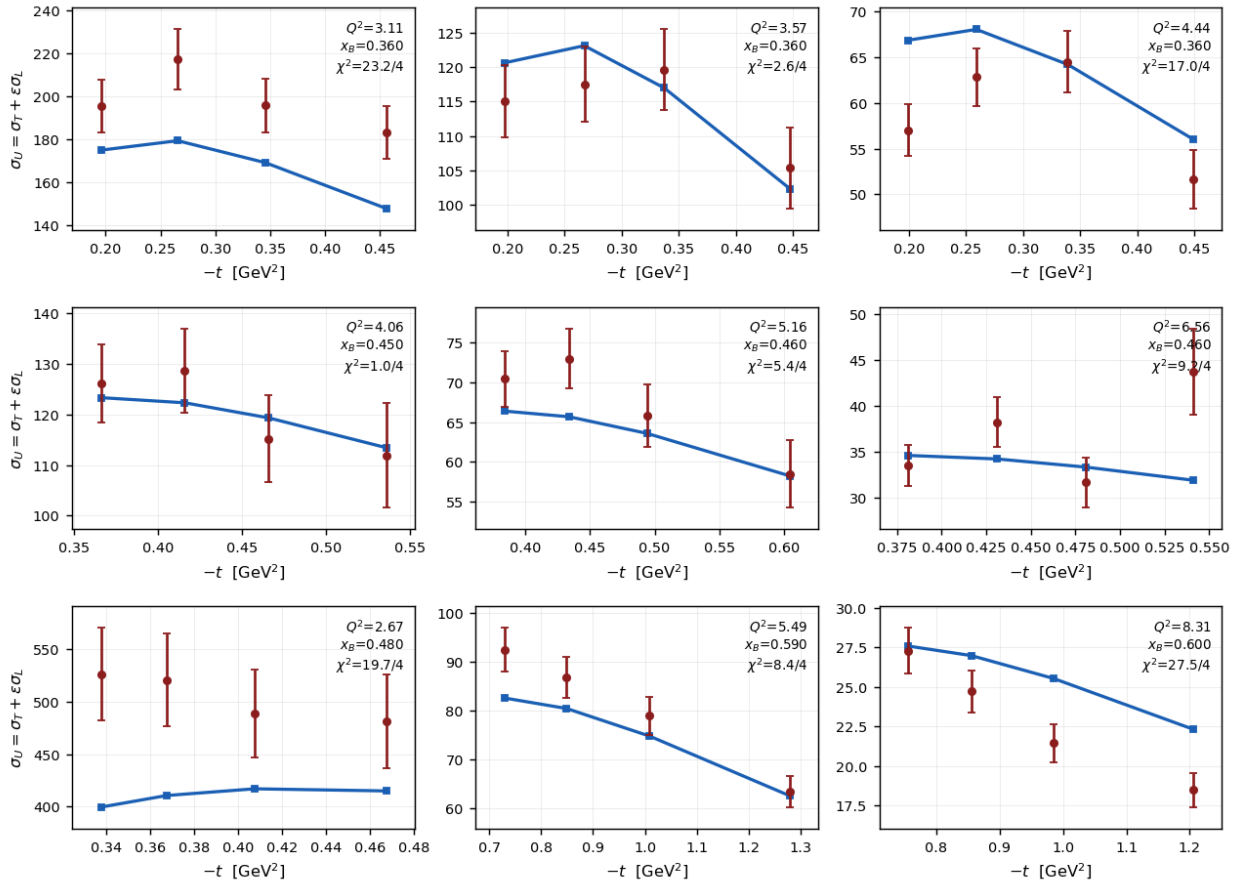
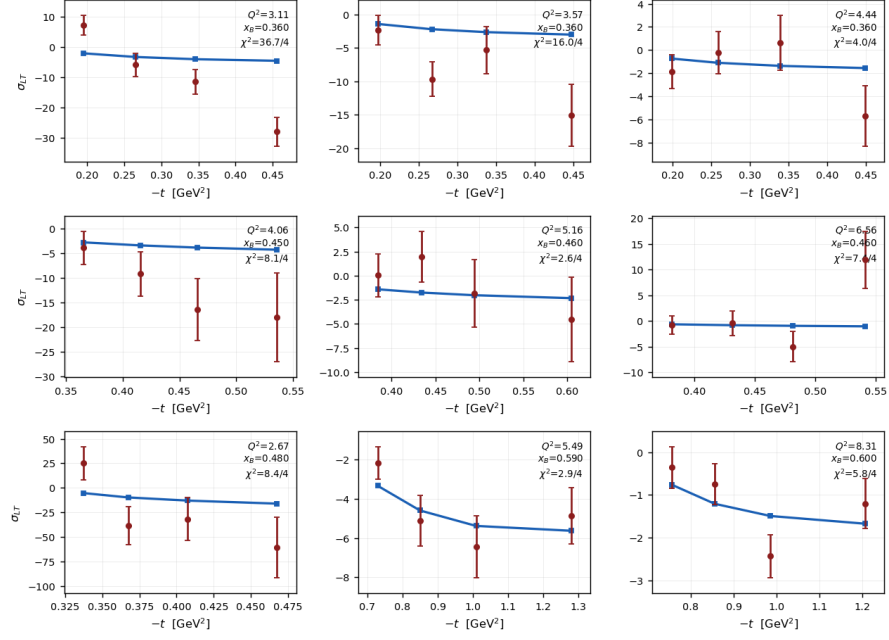


Figure 14: Hall A 2021, σ_U . The only data above $Q^2 = 4.3$, reaching 8.31 GeV²; left out of the fit the model overshoots these points by a factor 1.8.

Hall-A proton 12 GeV (2021) — σ_{LT} — IN THE FIT
 $\chi^2 = 91.9 / 36 = 2.55$ per point



Hall-A proton 12 GeV (2021) — σ_{TT} — IN THE FIT
 $\chi^2 = 102.3 / 36 = 2.84$ per point

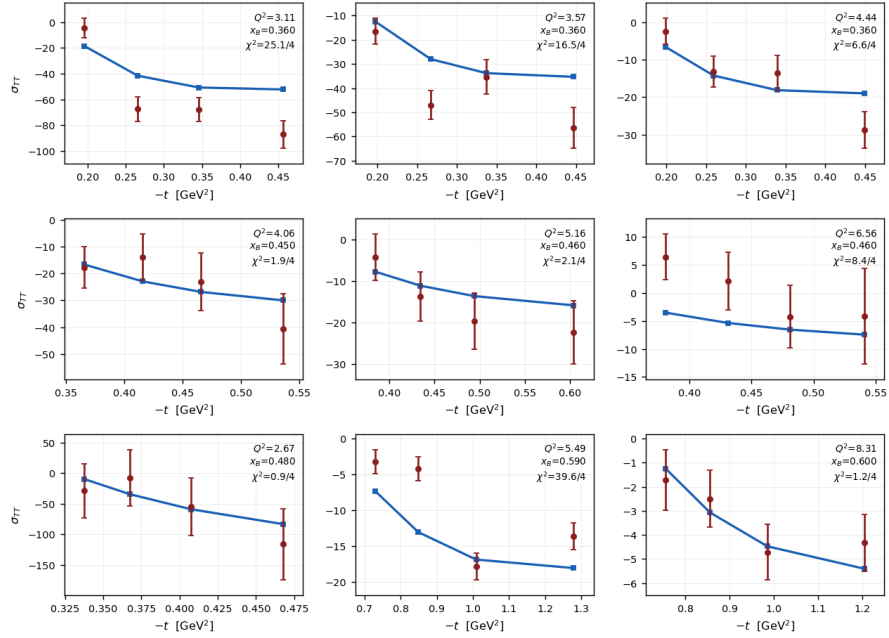


Figure 15: Hall A 2021, σ_{LT} and σ_{TT} .

Hall-A proton 12 GeV (2021) — $\sigma_{LT'}$ — IN THE FIT
 $\chi^2 = 50.0 / 36 = 1.39$ per point

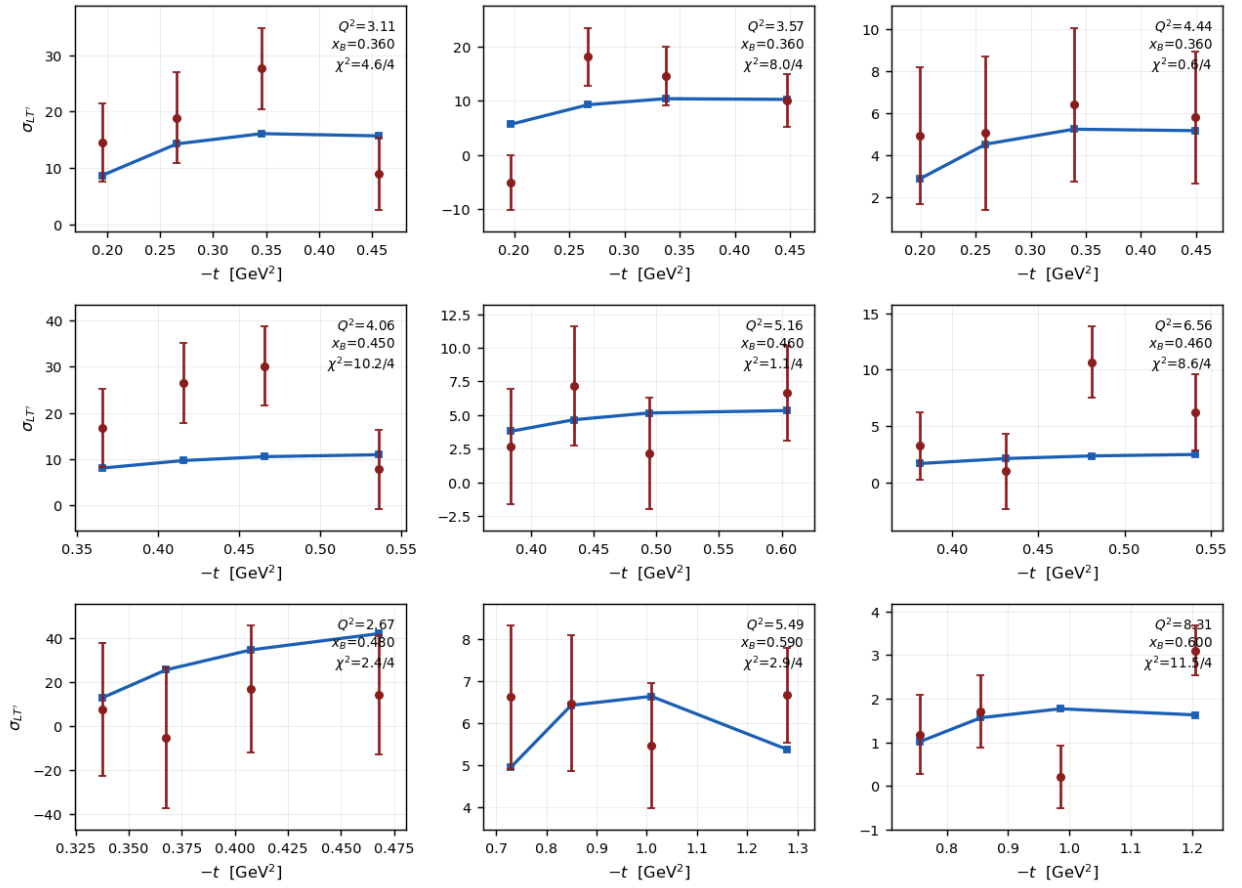


Figure 16: Hall A 2021, $\sigma_{LT'}$.

CLAS12 cross sections, preliminary — $\sigma_U = \sigma_U + \epsilon \sigma_U$ — NOT in the fit (blind prediction)
 $\chi^2 = 354.0 / 146 = 2.42$ per point

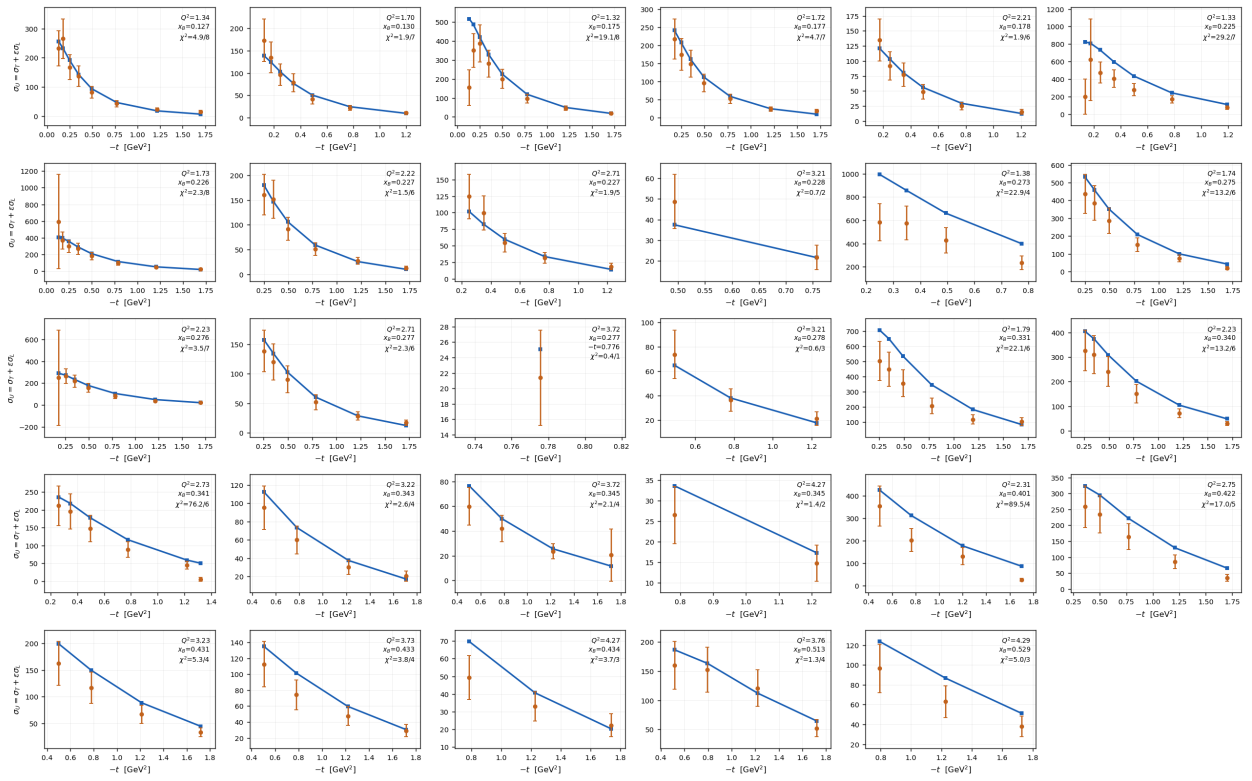


Figure 17: CLAS12 cross sections, σ_U — *not* in the fit, and preliminary. Predicted at 2.39 per point.

CLAS12 cross sections, preliminary σ_T — NOT in the fit (blind prediction)
 $\chi^2 = 2976.4 / 146 = 20.39$ per point

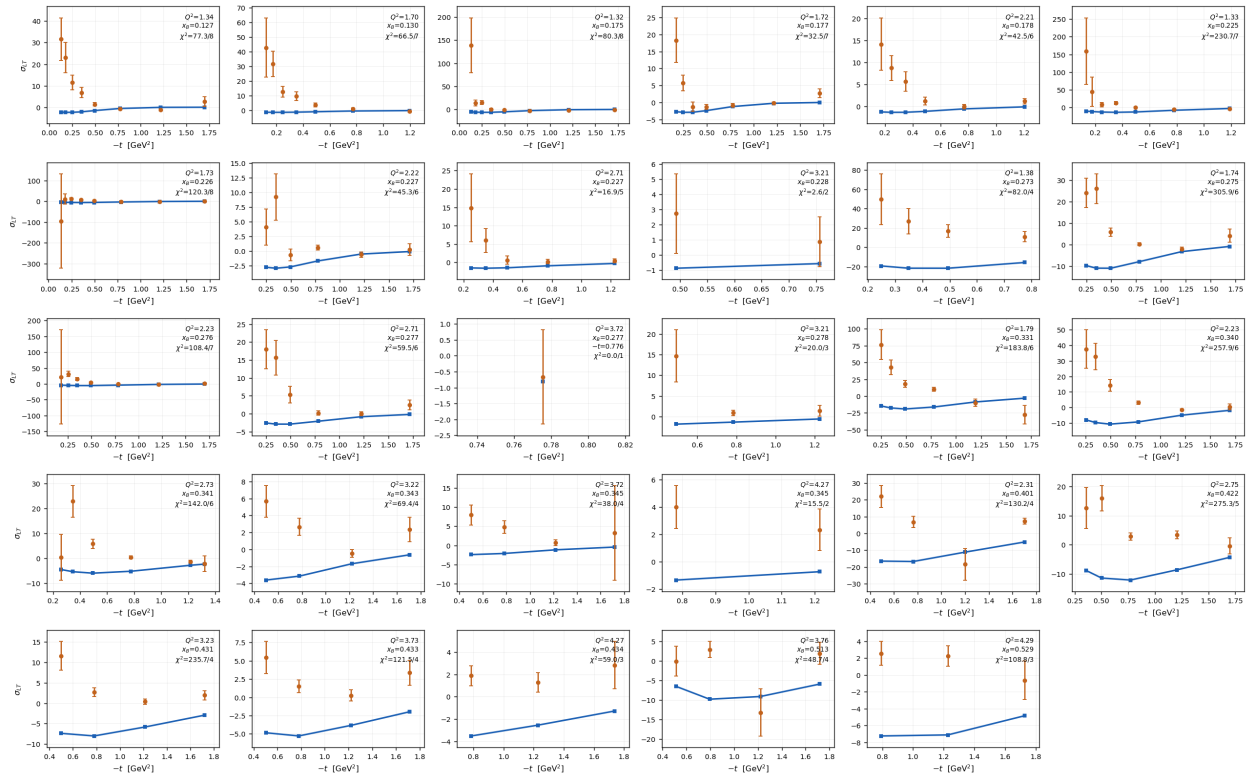


Figure 18: CLAS12, σ_{LT} , blind: 20.4 per point. This is the clearest single picture of the model's failure in σ_{LT} .

CLAS12 cross sections, preliminary — σ_{TT} — NOT in the fit (blind prediction)
 $\chi^2 = 522.9 / 146 = 3.58$ per point

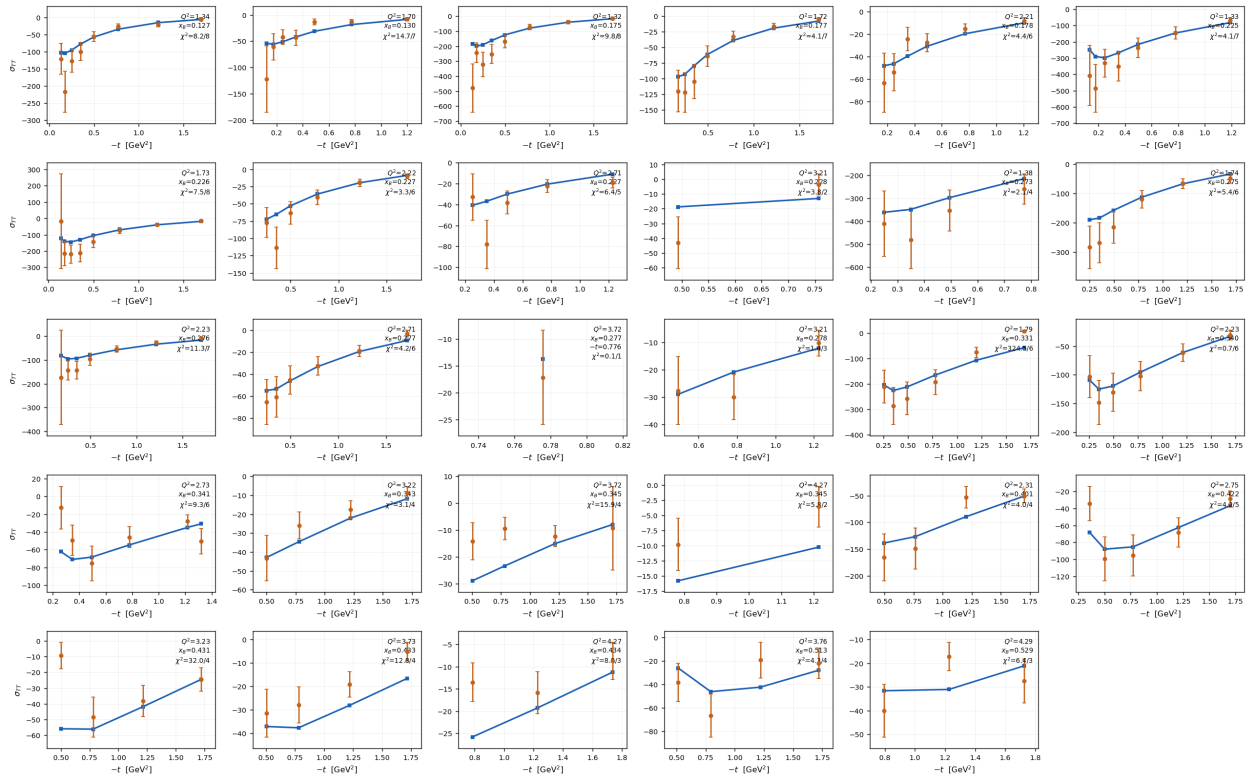


Figure 19: CLAS12, σ_{TT} , blind: 3.51 per point.

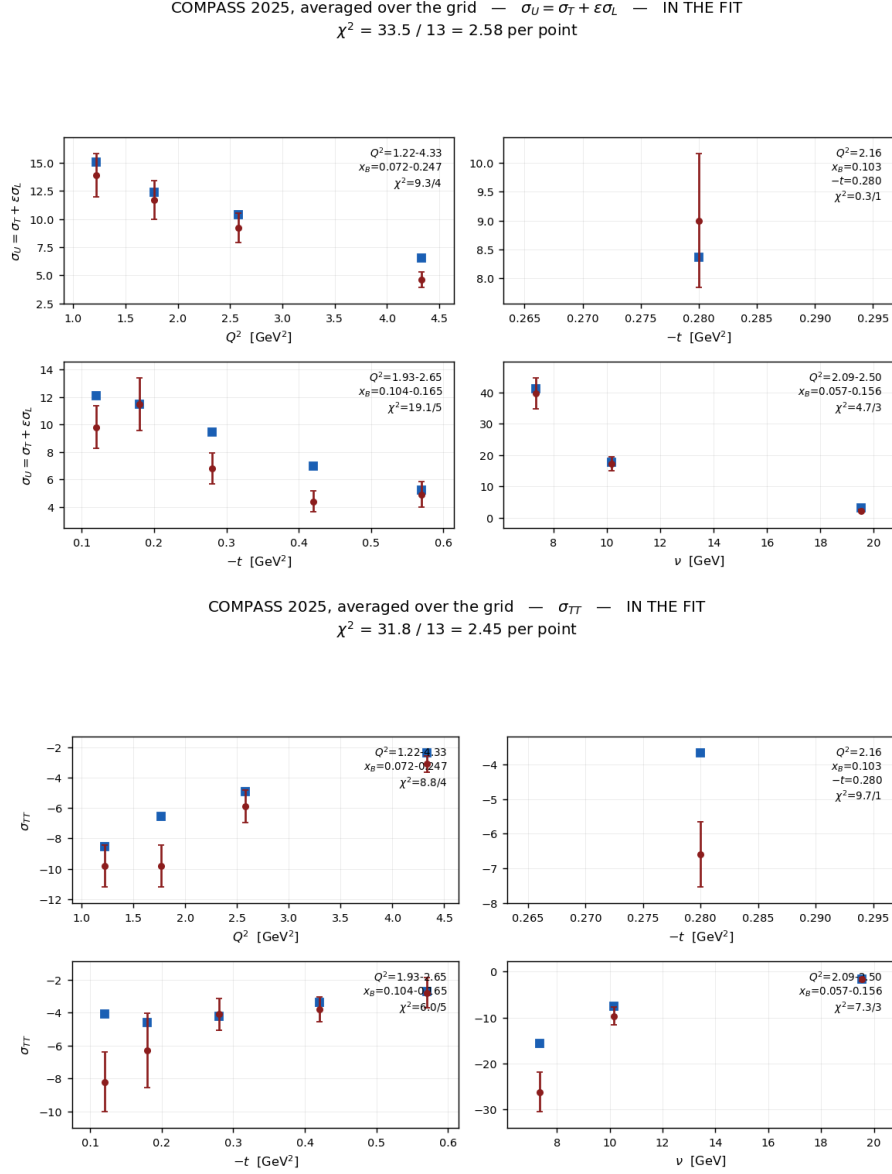


Figure 20: COMPASS 2025. The model is averaged over the four-dimensional grid of the measurement, eq. (16) of the paper, weighted by cell volume: a cell rectangular in (Q^2, ν) is not rectangular in x_B , and evaluating at the quoted mean kinematics overshoots by a factor five to eight.

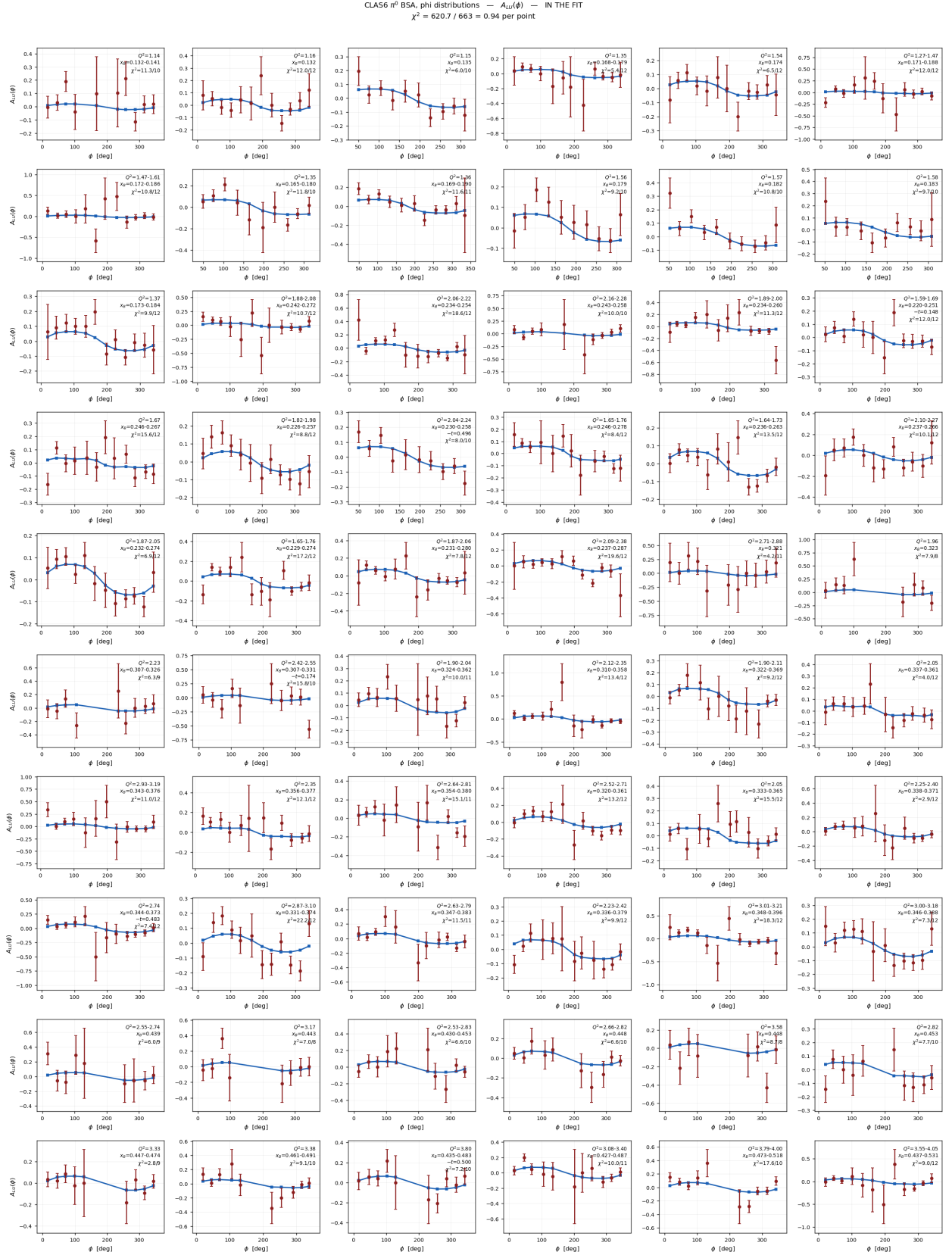


Figure 21: De Masi $A_{LU}(\phi)$: all 60 bins, 12 points in ϕ each. The model supplies the denominator of eq. (10) from its own σ_{LT} and σ_{TT} , so no free β, γ are fitted per bin — a three-parameter fit per bin is unstable, the denominator passing near zero in 13 of the 60.

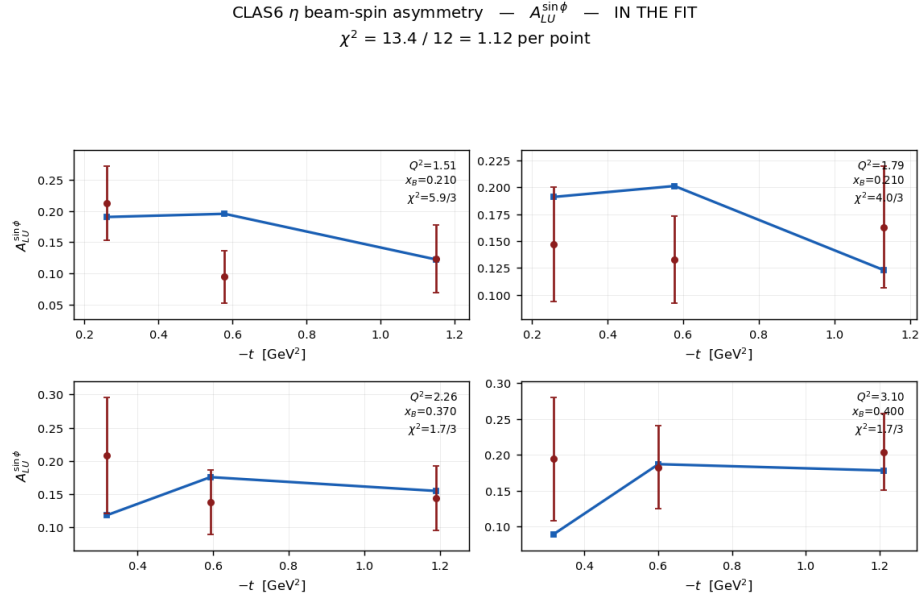
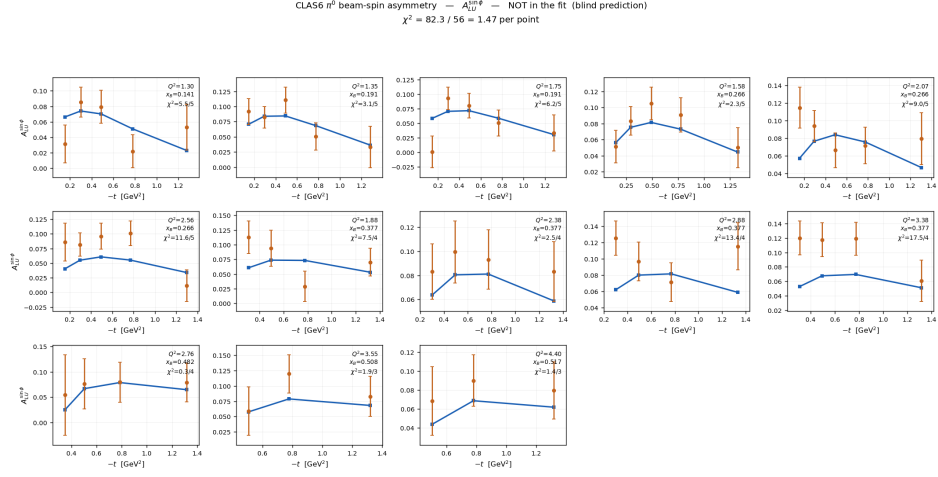


Figure 22: Top: the published De Masi α , kept out of the fit as the same measurement as the figure above. Bottom: the Zhao η beam-spin asymmetry, where the published GK curve is near zero.

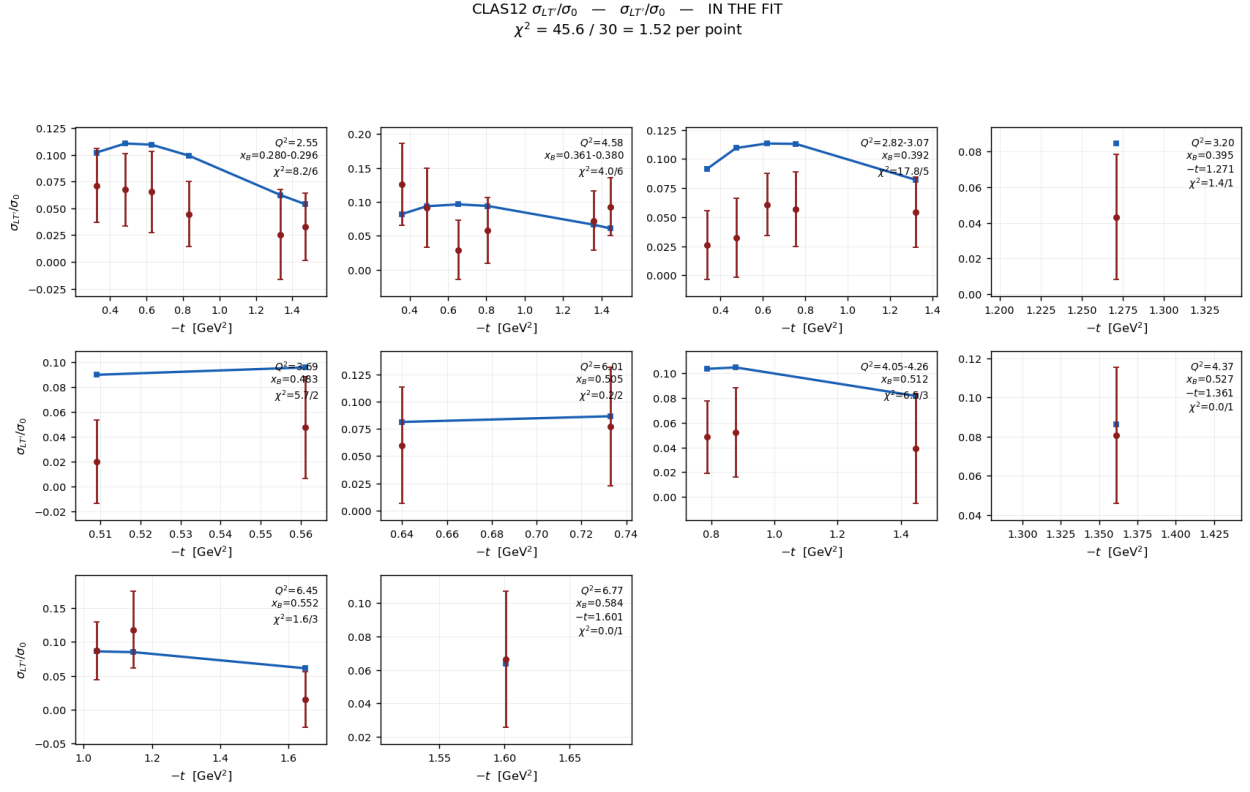


Figure 23: CLAS12 $\sigma_{LT'}/\sigma_0$.

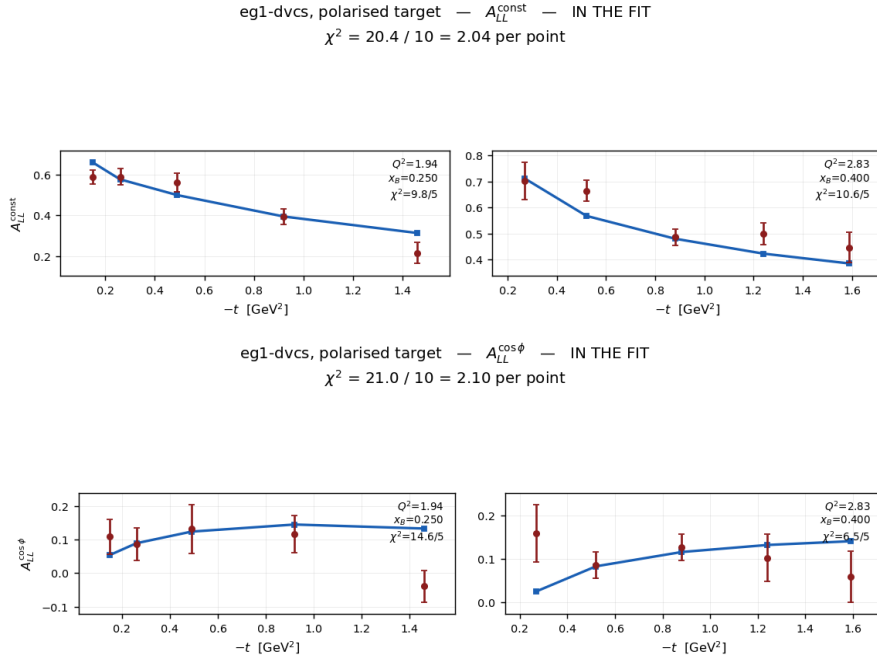
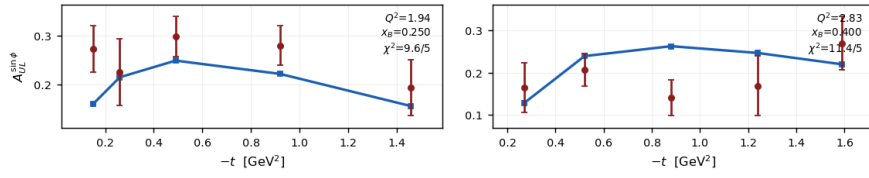


Figure 24: eg1-dvcs, polarised target: the A_{LL} moments, at the experiment's two settings. These are the only observables in the database sensitive to the relative phase of T_{01} and U_{01} ; without them it is unconstrained.

eg1-dvcs, polarised target — $A_{UL}^{\sin\phi}$ — IN THE FIT
 $\chi^2 = 21.0 / 10 = 2.10$ per point



eg1-dvcs, polarised target — $A_{UL}^{\sin 2\phi}$ — IN THE FIT
 $\chi^2 = 9.3 / 10 = 0.93$ per point

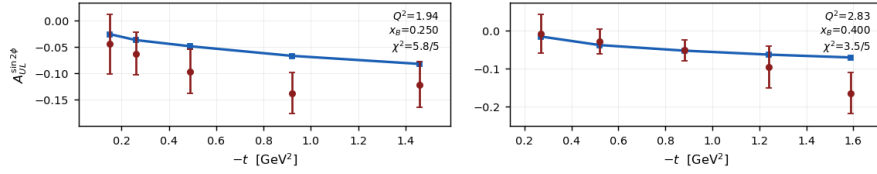


Figure 25: eg1-dvcs, the A_{UL} moments.

Reproducing this

Code, data and the full history are at https://github.com/vkubarovsky/rc_iter1. The fit shown is run `C1_with_compass`; its parameter vector is `runs/C1_with_compass/fitpar.npy` and every number in the tables above is generated from it by `tex/make_tables.py`, so nothing here is transcribed by hand. The data are `pi0_eta_database.xlsx`, md5 `04c3adf9d6e6b8d2b1b3729f4ff79d1d`; `export_db.py --check` verifies that the CSV cache the fit reads still matches the workbook.